

Investigating Acceptable Voice-based Notification Timings through Earable Devices: A Preliminary Field Study

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ABSTRACT

Earable devices, a subset of wearable technology, are designed to be worn on the ear and used in daily life. These innovative devices enable users to receive voice-based notifications through a minute built-in speaker without requiring any user operations, seamlessly integrating technology into everyday activities. The timing deemed acceptable for receiving voice-based notifications through earable devices varies based on the user and surrounding situation; thus, inappropriate notification timing may reduce usability. However, determining the safest and most comfortable timing for voice-based notifications using earable devices remains unclear. This study investigates the acceptable timing of voice-based notifications through earable devices. To explore the acceptable timing, we developed a smartphone application, SoNotify, which can send dummy voice-based notifications and collect sensor data on a smartphone and an earable device. Our field studies with eight participants showed that voice-based notifications were highly acceptable during outdoor walking, with an acceptance rate of approximately 86%. However, users tended to refuse notifications in situations in which they needed to concentrate on avoiding collisions with pedestrians, cyclists, or vehicles.

CCS CONCEPTS

• Human-centered computing → Ubiquitous and mobile computing systems and tools.

KEYWORDS

voice-based notification, earable device, interruptibility, attention management

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1 INTRODUCTION

The widespread use of mobile devices such as smartphones and music players has driven the evolution of earphones and headphones into lightweight, wireless accessories. These audio accessories have become commonplace, often worn continuously in daily life. For instance, approximately 86% of people have earphones, and 43% use them daily in the US [15]. Some smartphones are equipped with features that enable voice agents, such as Siri¹ or Google Assistant², to read notifications aloud, such as incoming calls and messages, when users wear headphones. This function represents voice-based notifications, which are alerts or messages delivered audibly. By contrast, screen-based notifications are perceived by reading the text displayed on the device screen. In the future, it is anticipated that people will extensively utilize voice-based notifications delivered through earable devices, facilitated by voice assistants or voice navigation systems.

Optimizing interruptions using digital devices is an important research topic for improving well-being in the current ubiquitous computing environment. According to an investigation, on average, individuals receive approximately 100 notifications per day from various applications on a smartphone [4]. Several methods have been proposed to solve this notification overload issue and optimize notification systems [2–5, 8, 9, 11–14]. For example, Okoshi et al. proposed a method [8, 9, 11, 12] to delay notifications until users reach opportune moments or situations using mobile sensing and machine learning, rather than delivering them immediately. Their evaluations in real-world, large-scale settings demonstrated their effectiveness in reducing users' cognitive load and mental effort,

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¹<https://www.apple.com/siri/> (Last Access 5/13/2024)

²<https://assistant.google.com/> (Last Access 5/13/2024)

shortening response times, and increasing click rates [10, 11]. Additionally, Cha et al. investigated the interruptible timing for interactions with smart speakers. They revealed that the key determinants of opportune moments are closely related to personal contextual factors, resource conflicts for dual-tasking, and the other contextual factors associated with the everyday routine[1]. Although effective systems for screen-based notifications and investigations focusing on voice interactions in indoor environments have been developed, no study has explored optimizing voice-based notifications through earable devices in outdoor situations.

Voice-based notifications are a type of notification provided through a smart device; this study specifically focuses on notifications delivered as voice messages through an earable device. Compared to the common screen-based notifications that have been researched in existing studies, voice-based notifications can be received hands-free and with an unobstructed field of view. Given recent concerns about smartphone addiction and distracted walking, voice-based notifications are considered an effective method for delivering notifications. However, they also have limitations, such as less flexibility in information acquisition and difficulty in pausing or adjusting the speed of notifications. Thus, the acceptable timing and utilization of voice-based notifications may differ from those of screen-based notifications.

User acceptance of voice-based notifications is significantly influenced by their environment and context. To explore this, we developed a smartphone application, SoNotify, which sends voice-based notifications and collects both user acceptance and sensor data from smartphones and earable devices. Additionally, we conducted a preliminary field study, collected data, and identified key influencing factors.

The main contributions of this study are as follows:

- This is the first study to investigate acceptable timing for voice-based notifications.
- We developed SoNotify, a smartphone application that can send voice-based notifications and collect sensor data from smartphones and earable devices with their feedback.
- The preliminary field study revealed that avoiding collisions with other pedestrians, bikes, and vehicles is a key factor in not accepting voice-based notifications.

2 GOAL AND APPROACH

We aimed to investigate acceptable timings for voice-based notifications and evaluate these timings using sensor data. We collected sensor data such as acceleration and location information from smartphones and earable devices to analyze the relationship between the acceptance of notifications and user circumstances. In the future, we aim to detect acceptable timings using sensor data and develop an optimal voice-based notification system. To achieve this, we developed SoNotify and conducted a preliminary field study.

2.1 Data Collection Method –SoNotify

SoNotify³ is an iOS application designed to collect sensor data and user feedback regarding the acceptance of voice-based notifications. Fig. 1 illustrates the system architecture of SoNotify. The MySQL database was implemented as a data collection server on the Google

³SoNotify is a portmanteau of Sonority and Notify.

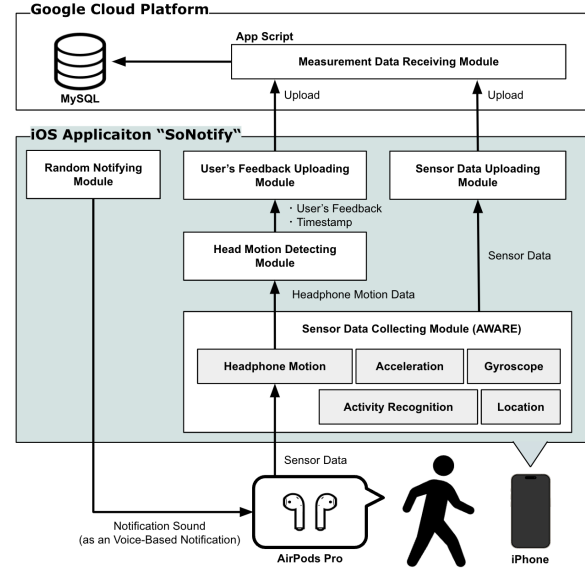


Figure 1: System architecture of SoNotify.

Cloud Platform⁴. Fig. 2 illustrates the user experience with SoNotify. (1) Tapping the start button activates the sensors and initiates the collection of sensor data from the iPhone and AirPods Pro using the AWARE framework [6, 7]. AWARE is an iOS library gathering various types of sensor data, such as accelerometer, gyroscope, battery, network, and location information, which are embedded in an iPhone⁵, AppleWatch⁶, and AirPods⁷. The SoNotify system collects a wide range of sensor data from both iPhones and AirPods Pro. For an iPhone, these data include accelerometer, gyroscope, activity recognition, and location data, which are collected at a frequency of 100 Hz for accelerometric and gyroscopic data, 0.5 Hz for activity recognition, and 1 Hz for location data. Sensor data collected from AirPods Pro – consisting of an accelerometer, gyroscope, and gravity – were sampled at a rate of 20 Hz. (2) Notification sounds are sent randomly indicating the arrival of voice-based notifications. (3) Users should provide feedback on whether they can accept voice-based notifications at the moment using head motions. When interacting with a voice agent through earable devices, users can respond to feedback or take subsequent action using their voice. However, in quiet situations, such as on a train or in a library, users cannot use voice feedback. Therefore, users are expected to answer by their head motion. They accept voice-based notifications by nodding their heads or decline by shaking their heads. A head movement classification model is developed using CreateML⁸ and takes as input the rotation rate (x, y, z) of the headphone motion. A corresponding feedback tone was played to confirm that the system recognized the intended head movements of the participants. (4) Timestamp when the notification sound is sent and user feedback

⁴<https://cloud.google.com/gcp?hl=en> (Last Access 5/13/2024)

⁵<https://www.apple.com/iphone/> (Last Access 5/13/2024)

⁶<https://www.apple.com/watch/> (Last Access 5/13/2024)

⁷<https://www.apple.com/airpods/> (Last Access 5/13/2024)

⁸<https://developer.apple.com/machine-learning/create-ml/> (Last Access 5/13/2024)

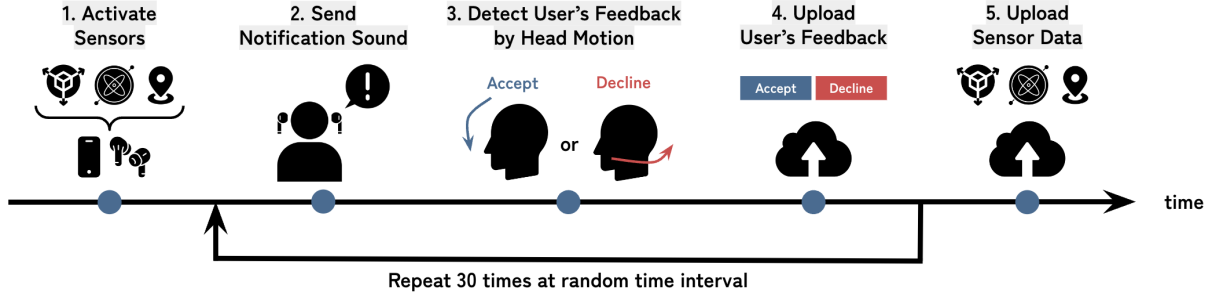


Figure 2: Field study protocol with SoNotify: Data collection and user manipulation overview.

is uploaded to the server. (5) Upon completion of 30 notification sessions, the sensor data were uploaded to a server.

2.2 Field Study Protocol

We recruited 10 participants (aged 20–24, five males and five females) who used earable devices in their daily lives. Owing to data collection problems with two participants, this study reports the results obtained from eight participants. The experiment used an iPhone (iOS 17.4.1) with SoNotify installed, AirPods Pro (2nd generation), and a wearable camera (Insta360 Go3) to record the experimental conditions.

In this field study, we focused on outdoor walking situations as preliminary research. The study was conducted in three urban areas in Tokyo, Japan. We hypothesized that the ability to receive voice-based notifications while walking outdoors is significantly influenced by the presence of other pedestrians and vehicles. Therefore, each route was designed to include at least six intersections with traffic signals. Throughout the experiment, the participants received 30 dummy notification sounds at random intervals while the system collected sensor data from iPhone and AirPods Pro. They were instructed to indicate their readiness to receive a voice-based notification by moving their head upon hearing the notification sound. The sensor data and user's feedback were uploaded to the server. After data collection, we conducted semi-structured interviews to explore the reasons behind the participants' acceptance or rejection of voice-based notifications and their preferences regarding the usage of such notifications. To ensure participant safety, we set the AirPods to transparency mode and had assigned the experimenter followed to monitor them in case of any danger. The Research Ethics Committee of the Center for Spatial Information Science at the University of Tokyo approved this study.

3 RESULT

Table 1 lists the acceptance rates of voice-based notifications in relation to the walking states. All stopping situations refer to when participants stop to wait for a traffic light. Approximately 85% of the notifications were accepted while walking, and approximately 96% were accepted while stopping. The high acceptance rate of notifications while walking underscores the advantages of voice-based notifications such as hands-free operation and unobstructed vision. Fig. 3 presents the accepted and declined points on the maps of the three areas. In this experiment, the participants walked on

Table 1: Acceptance rate of the voice-based notifications with each moving state.

Moving State	Accept	Decline	Acceptance Rate
Walking	180	32	0.85
Stopping	27	1	0.96
Total	207	33	0.86

Table 2: Acceptance rate of the voice-based notifications with different road types in walking conditions.

Road Type	Accept	Decline	Acceptance Rate
Sidewalk	138	11	0.93
Roadside	34	12	0.74
Crosswalk	8	9	0.47
Total	180	32	0.85

three different types of roads: (1) sidewalks, where the roadway and sidewalk are physically separated by guardrails, planting strips, or steps. (2) Roadside, where the roadway and sidewalk are not physically separated, but are delineated by white lines. (3) A crosswalk is a marked area on a road that pedestrians can safely cross. Acceptance rates were the highest on sidewalks, with 93% of the notifications accepted. However, the acceptance rates decreased to 74% on narrow roadsides, and further decreased to 47% at crosswalks.

Fig. 4 shows the reasons for declining voice-based notifications. The most common reason, accounting for approximately 80%, is "to avoid collisions with others." Others refer to vehicles, bicycles, and other pedestrians. Voice-based notifications were typically rejected when individuals concentrated on being cautious of other traffic and other pedestrians to walk safely. In addition, they were declined at poorly visible intersections because of the possibility of cars suddenly appearing or voice-based notifications becoming inaudible from ambient noise. Although the situations observed in this experiment were limited, it can be inferred that voice-based notifications should not be sent when pedestrians are attempting to avoid collisions with other traffic.

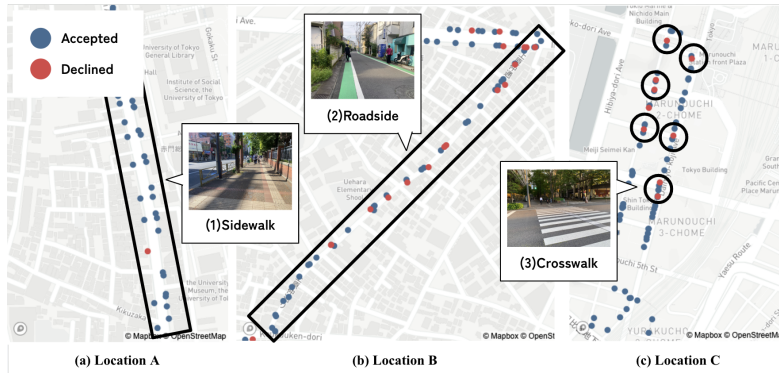


Figure 3: Spatial visualization of accept and decline location for notification sounds and three road types

4 DISCUSSION

To further enhance the effectiveness of voice-based notifications in the future, it is crucial to accurately predict acceptable timing. In the semi-structured interviews, most participants commented that they wanted to use voice-based notifications for more important and urgent notifications than screen-based ones. Considering these preferences, it appears that avoiding unacceptable moments rather than delaying notifications to an opportune moment is suitable for optimizing voice-based notification timings.

This preliminary field study has several limitations. First, the experiment was conducted in controlled scenarios, which is insufficient for exploring the optimization of notifications in daily life. While we identified "avoiding collisions with others" as a key contextual factor for optimizing voice-based notifications, other contextual factors are likely to emerge in large-scale and long-term experiments. Second, optimizing voice-based notifications requires consideration beyond timing. Factors to consider include the type of information included in the notifications, suitable notification sounds to recognize through earable devices, and notifications interfering with other audio media the users are listening to.

In the future, we plan to conduct large-scale and long-term data collection experiments. Based on the contextual key factors identified in these experiments, we aim to construct a prediction model for acceptable voice-based notification timings. When developing a practical system, privacy considerations are crucial. Analyzing a user's situation or context using cameras, microphones, and location data can be effective; however, these types of data are highly personal. Therefore, it is essential to carefully consider the creation of the prediction model using privacy-friendly sensor data. Furthermore, developing an optimal notification system that assumes a multi-device environment, such as combining smartphones, smartwatches, and earphones, is also crucial.

5 CONCLUSION

This study primarily aimed to investigate the timing and factors affecting the acceptance of voice-based notifications using earable devices. To this end, we developed an iOS application, SoNotify. This application delivers voice-based notifications and collects multiple sensor data, including location, accelerometer, and gyroscope,

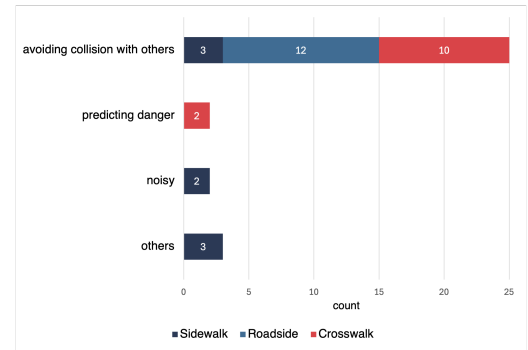


Figure 4: Reasons for declining voice-based notifications and road attributes during responses

from a smartphone and an earable device. We conducted field experiments in walking scenarios through three distinct areas in Tokyo, Japan. Data analysis from eight participants, amounting to 240 voice-based notifications, revealed that approximately 96% notifications were accepted when participants stopped at intersections and approximately 85% when they were walking. The ability to receive notifications while walking is a typical feature of voice-based notifications. However, these notifications were often declined when participants needed to avoid potential collisions with objects or individuals such as cars, bicycles, or pedestrians. To ensure the safe use of voice-based notifications, situations in which notifications should not be sent must be detected. In the future, we plan to conduct large-scale and long-term experiment to further explore acceptable voice-based notification timings and develop a prediction model using sensor data. This knowledge will be valuable for developing an optimal voice-based notification system.

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